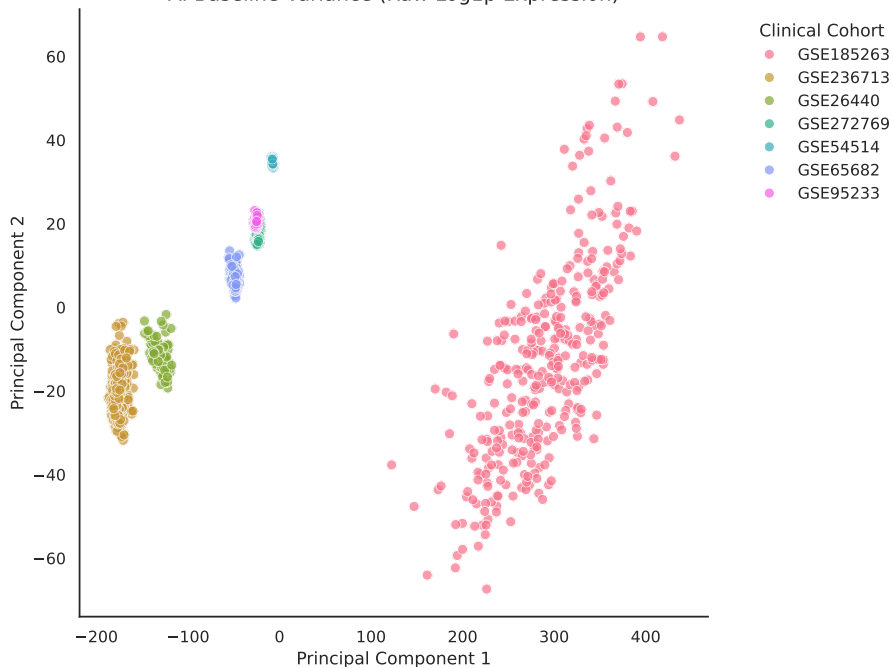
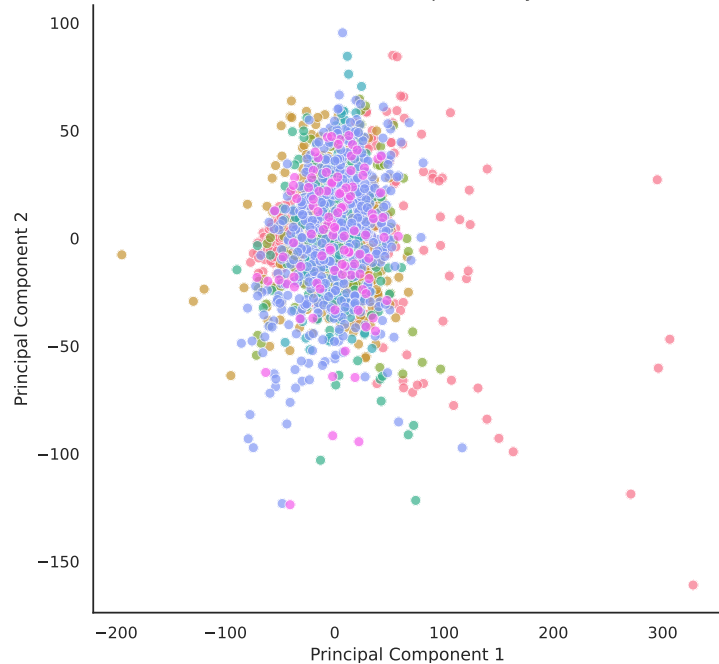


A. Baseline Variance (Raw Log1p Expression)



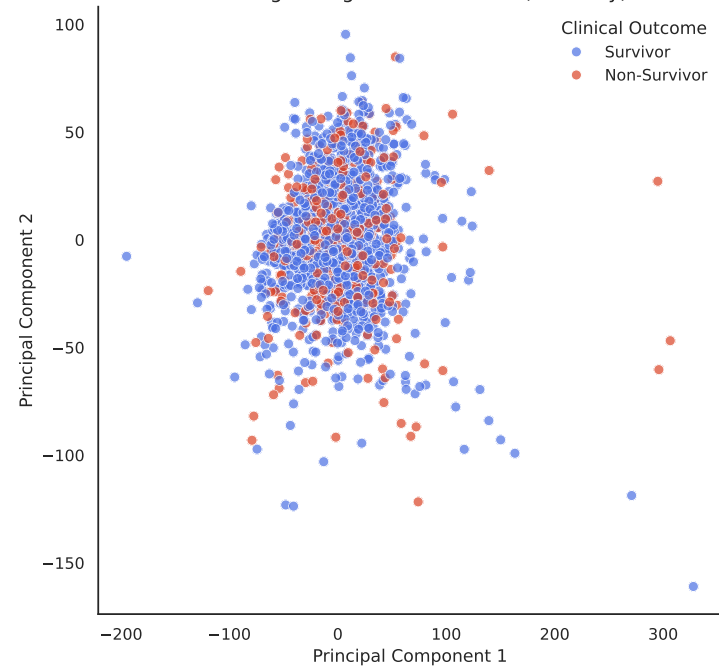
B. Post-Harmonization (Empirical Bayes ComBat)



C. Technical Mixing by Cohort



D. Biological Signal Preservation (Mortality)



Statistical Validation (ANOVA)

- Technical Batch (Cohort): PC1-PC3 $p = 1.000$ (Variance Neutralized)
- Biological Signal (Mortality): PC3 $F = 11.88$, $p < 0.001$ (Signal Preserved)